Sabatino Santangelo

Planetary Scientist | Aerospace Engineer

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Born July 10, 2000, Naples, Italy

Summary

Space engineer with a scientific approach to problem solving and a strong capacity for independent learning, currently working on lunar science and exploration. Goal-oriented, adaptable, and passionate about astrophysics and science communication. Eager to contribute to space exploration.

Education & research

Graduate Researcher | Planetary Geodynamics
German Aerospace Center (DLR), Institute of Space Research — Nov. 2024 – Present

- Investigating **lunar asymmetry** through **thermal evolution** modeling using finite-volume methods in C++. Data assimilation approach based on global models, incorporating results of **crustal thickness**, **tidal response**, and **magma ocean crystallization** modeling, alongside geospatial and geochemical datasets.
- Secondary focus in lunar **heat-flow measurements** and develop model-based science cases to support **mission planning**.

Master's in Aerospace Engineering
Delft University of Technology — Sep. 2022 – Oct. 2024

Focus on Planetary Exploration. **Grade:** 8.5/10.

- **Key courses:** Concurrent mission design, planetary sciences, rocket propulsion, numerical astrodynamics, re-entry, thermal control, and systems engineering.
- **Thesis:** "Interior asymmetry effects on the thermal evolution of the Moon."
  Studied effects of asymmetric crustal structure and radiogenic heating distribution in geodynamic models.

Bachelor's in Aerospace Engineering
Federico II University of Naples — Sep. 2019 – Jul. 2022

Thesis: "RANSAC-based algorithm for pose determination of cooperative satellite."
Grade: 110/110 cum laude (top 5% of the program).

Work experience & projects

Lunar Multi Sensor Station (LMSS) Concept Study
Mission Scientist — Apr. 2026 – Present

Contribute to project management and payload design for a DLR-led Phase-0 study of a multipurpose lunar relay station deployable by astronauts.

ESA – Moonraker Proposed Mission (Phase A)
Supporting Scientist — Jun. 2026 – Present

Support the science principal investigator at DLR in defining the science case, developing and reviewing requirements, and maintaining requirements traceability.

ESA – Outreach Activities
Science Communicator — Apr. 2026 – Present

Deliver interactive science presentations in German for children and families at ESA open days and space exhibition events. Engage visitors at exhibition stands through accessible explanations and discussion.

Research Internship
German Aerospace Center (DLR), Institute of Space Research — Sep. 2023 – Dec. 2023

Studied the Moon's thermochemical evolution by coupling 3D lunar evolution models with crustal thickness models. Validated results against mission data, particularly Apollo 15 and 17 heat-flow measurements.

List of publications

Journal articles

Santangelo, S., Plesa, A.-C., Broquet, A., Breuer, D., & Grott, M. (2025).
"Present-day thermal state and surface heat flux of the Moon."
Journal of Geophysical Research: Planets, *130*(12), e2025JE009458. doi:10.1029/2025JE009458

Review chapters

Breuer, D., Plesa, A.-C., Broquet, A., Bonnet Gibet, V., Briaud, A., Cascioli, G., Grott, M., Hußmann, H., Knapmeyer, M., Oberst, J., **Santangelo, S.**, & Schwinger, S. (2026).
"Lunar asymmetry and interior dynamics."
In *Lunar Exploration*, *Advances in Geophysics*, Vol. 67, pp. 55–117. Elsevier. doi:10.1016/bs.agph.2026.06.002

Additional projects

ESA – Concurrent Engineering Workshop
Payload Subsystem Team Leader — Sep. 2025

Defined the science case and selected the payload for a Phase-0 study of a Jupiter flyby mission using concurrent engineering.

Micro-Resistojet Design Project
Thruster Modeling & Preliminary Design — Jun. 2023 – Aug. 2023

Developed a software tool to model water-propelled Vaporizing Liquid Micro-Resistojet (VLM) performance. Iterated designs and evaluated trade-offs to optimize performance against mission requirements.

Moon Industry Student Team
Chief Engineer — Mar. 2023 – Sep. 2023

Managed, coordinated, and reviewed the engineering work of a **15-student team** tasked with designing, building, and testing a lunar mining twin-rover system for NASA's "Break the Ice – Lunar Challenge."

Polar Solar Orbiter Concurrent Design Project
Systems & Payload Engineer — Feb. 2023 – Apr. 2023

Contributed to the concurrent design of a solar-sailing polar solar orbiter. Managed system budgets, integration, and requirements compliance; defined scientific goals, selected the payload, and supported instrument accommodation.

Pluto Lander Mission Design
Science Case & Payload Design — Nov. 2022 – Jan. 2023

Defined scientific goals and designed the payload for a hopping lander concept in the Pluto–Charon system. Studied Pluto's geology and explored innovative in situ measurement strategies to meet mission objectives.

Digital skills

Python (advanced); MATLAB (intermediate); C++ (intermediate); LaTeX (advanced).

Languages

Italian (native); Spanish (fluent); German (intermediate); English (proficient).
Cambridge CPE certificate, 2021.